that govern their behavior. Particle-wave duality is an easy extension of the things you've already learned, as we'll see in this lecture. But before we get into the physics, I want to review some mathematics, some of which is old —it appeared in earlier lectures—and some of which is new.

8.1 MATHEMATICAL INTERLUDE: WORKING WITH CONTINUOUS FUNCTIONS

8.1.1 Wave Function Review

We'll be using the language of wave functions in this lecture, so let's review some of that material before we dive in. In Lecture 5, we discussed wave functions as abstract objects, without explaining what they had to do with either waves or functions. Before correcting this omission, I will review what we discussed earlier.

Begin by picking an observable **L**, with eigenvalues λ and eigenvectors |λ>. Let |Ψ> be a state-vector. Since the eigenvectors of a Hermitian operator form a complete orthonormal basis, the vector |Ψ> can be expanded as

$$|\Psi\rangle = \sum_{\lambda} \psi(\lambda)\ |\lambda\rangle. \qquad (8.1)$$

As you recall from Sections 5.1.2 and 5.1.3, the quantities

$$\psi(\lambda)$$

are called the wave function of the system. But notice: the specific form of $\psi(\lambda)$ depends on the specific observable **L** that we initially choose. If we pick a different observable, the wave function (along with the basis vectors and eigenvalues) will be different, *even though we're still talking about the same state*. Therefore, we should qualify the statement that $\psi(\lambda)$ is the wave function associated with |Ψ>. To be more precise, we should say that $\psi(\lambda)$ is